O'REILLY ARCHITECTURAL KATAS 2026

Von Digitalis Estates

An estate where every AI model is optional, replaceable and continuously scored.

Team Ground Truth

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THE BRIEF

The Countess has inherited a sprawling estate

40

historic 18th-century rides

200+

exotic and poisonous animals

55

enclosures, aquatic and land-based

5,000

visitors a day today

What was asked for

1. Sell tickets, including family passes
2. Know which parts of the estate are popular, to invest and deploy staff
3. Monitor animal health, feeding and the jumping piranha population
4. Grow visitor numbers and make the estate profitable

Target: 15,000 visitors a day within three years

THE CHALLENGE

Three problems named, and what's at stake



No idea what's popular

Investment and staffing are guesswork. Busy is not the same as valuable: a corridor near the entrance looks popular.



Sick animals cost more

Care is costly, and far more so once an animal is ill. Many of the 200 animals are venomous, so handling them is a risk in itself.



Visitors don't come back

The estate needs returning visitors, and nobody knows which experiences bring people back.



Fail, and the family may be forced to sell their carnivorous plant collection

CONSTRAINTS

What we had to design within

Business constraints

Profit, not visitor count

Growth that loses money per head makes things worse

Safe and insurable, always

Poisonous animals, carnivorous plants and 18th-century rides among growing crowds

Animal care is costly

Far more so once an animal falls ill

Returning visitors matter

Growth has to come from people choosing to come back

Technical constraints

Patchy Wi-Fi

Coverage across the park is unreliable; outages are normal

Cloud is allowed, if data gets there

Data must reliably reach the cloud despite the network

A budget for MQTT devices

Sensors and gateways on a fixed hardware budget

AI changes every few months

The best model, its price and its provider may all change

PROJECTED IMPACT

What the estate stands to gain

+36%

revenue per visitor, from $22 to $30

$12M

a year at today's 5,000 visitors, before any growth

3×

visitors by year 3, inside a 7,500 safety ceiling

~4×

annual revenue by year 3, about $33M to $135M

Visitor growth needs capital, capacity and years. Spend per visitor needs no new rides: every extra $1 a visitor spends is worth $1.5M a year.

How we'd know

Each projection names its measure: zone-level P&L, gate admissions, rostered hours. Assumptions are stated in our README, sections 3 and 14.

BUSINESS MODEL

Where the money comes from



Adventure Park

Price the 40 rides as heritage: engineering tours, reserved slots, evening operation



Zoo and Aquarium

Keeper experiences, adopt-an-animal, nocturnal tours, weekday school programmes



Gardens

The collection she'd have to sell becomes stock: propagation, workshops, photography permits



The estate

Venue hire, film locations, seasonal night events and the exploding-gnome brand



A zone-level P&L, with no AI at all

Revenue less allocated cost, per zone, every day. Popularity is the wrong question; contribution is the right one.



Guardrails on revenue

Paid animal encounters are welfare-gated. Capacity ceilings are set by safety, never by demand.

OUR QUESTION

How might we use AI only where it earns its place, to make the estate safe, well-run and profitable on a network that can't be trusted?

Our thesis: Ground Truth

Every AI capability is paired with a mechanism that produces ground truth to score it against.

Our rule

Where ground truth is expensive to get, we either don't use AI, or we don't let it decide.

THE TEAM

Team Ground Truth, from Zemoso Technologies



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Senior Tech Lead



Ranjith DVL

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Sourav Pujara

Principal Software Engineer



Sanket Kisan Raut

Principal Software Engineer



Mohammad Faizan Haidery

Senior DevOps Engineer

We drafted in parallel and reviewed together: every use case, diagram and ADR passed the whole team. Product, architecture, engineering and DevOps, one table.

HOW WE WORKED

We started with people, not technology

9
actors, event-stormed action by action

→

25
aggregates from the events they produce

→

6
architectural quanta, each with its own needs

→

5 + 3
use cases, plus three that span them all

Actor	Action	Event	Aggregate
Visitor	Create family pass	Ticket created	Ticket
Booking admin	Check pass validity	Pass validated	Ticket
Caretaker	Report health information	Health info updated	HealthRecord
Rides operator	Close ride	Ride closed	Ride
AI system	Escalate to a person	Approval sent	—
Evals	Evaluate AI outputs	Corrections identified	AI Model

What it changed

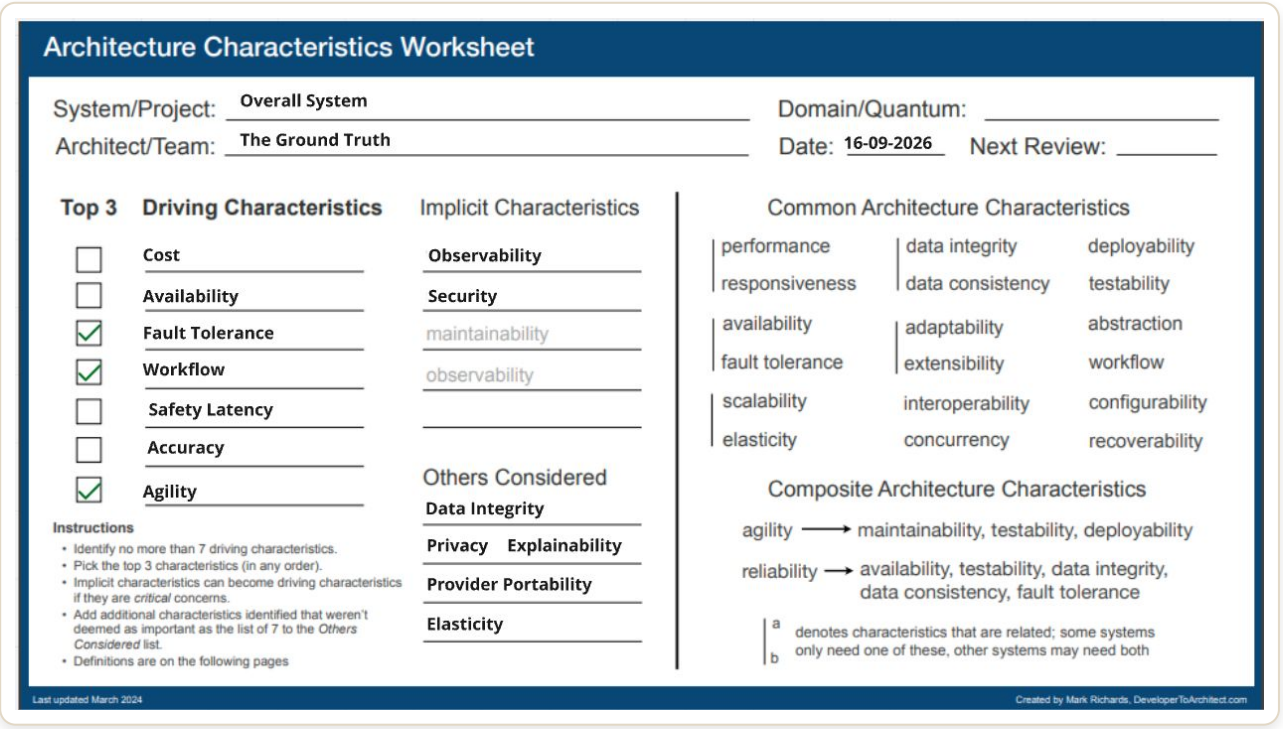
The AI system is an actor

Beside the vet on the board, every AI action had to end in a person's approval or a recorded evaluation.

Keepers record offline

The actor work, not the brief, showed care takers need to capture notes without signal or forms.

We chose what to optimise for before adding any AI



Architecture Characteristics Worksheet

System/Project: Overall System
Domain/Quantum:
Architect/Team: The Ground Truth
Date: 16-09-2026 Next Review:

Top 3	Driving Characteristics	Implicit Characteristics
☐	Cost	Observability
☐	Availability	Security
☑	Fault Tolerance	maintainability
☑	Workflow	observability
☐	Safety Latency	
☐	Accuracy	
☑	Agility	

Instructions

- Identify no more than 7 driving characteristics.
- Pick the top 3 characteristics (in any order).
- Implicit characteristics can become driving characteristics if they are *critical* concerns.
- Add additional characteristics identified that weren't deemed as important as the list of 7 to the *Others Considered* list.
- Definitions are on the following pages

Others Considered

Data Integrity
Privacy Explainability
Provider Portability
Elasticity

Common Architecture Characteristics

performance	data integrity	deployability
responsiveness	data consistency	testability
availability	adaptability	abstraction
fault tolerance	extensibility	workflow
scalability	interoperability	configurability
elasticity	concurrency	recoverability

Composite Architecture Characteristics

agility → maintainability, testability, deployability

reliability → availability, testability, data integrity, data consistency, fault tolerance

[a] [b] denotes characteristics that are related; some systems only need one of these, other systems may need both

Last updated March 2024 — Created by Mark Richards, DeveloperToArchitect.com

Mark Richards' worksheet, completed for the whole estate before any AI decision was made.

Top three driving characteristics

Fault tolerance

The Wi-Fi drops; admission and alarms must carry on

Workflow

Every job spans devices, retries and partial failure

Agility

AI changes every few months; the estate's core must not

Estate platform

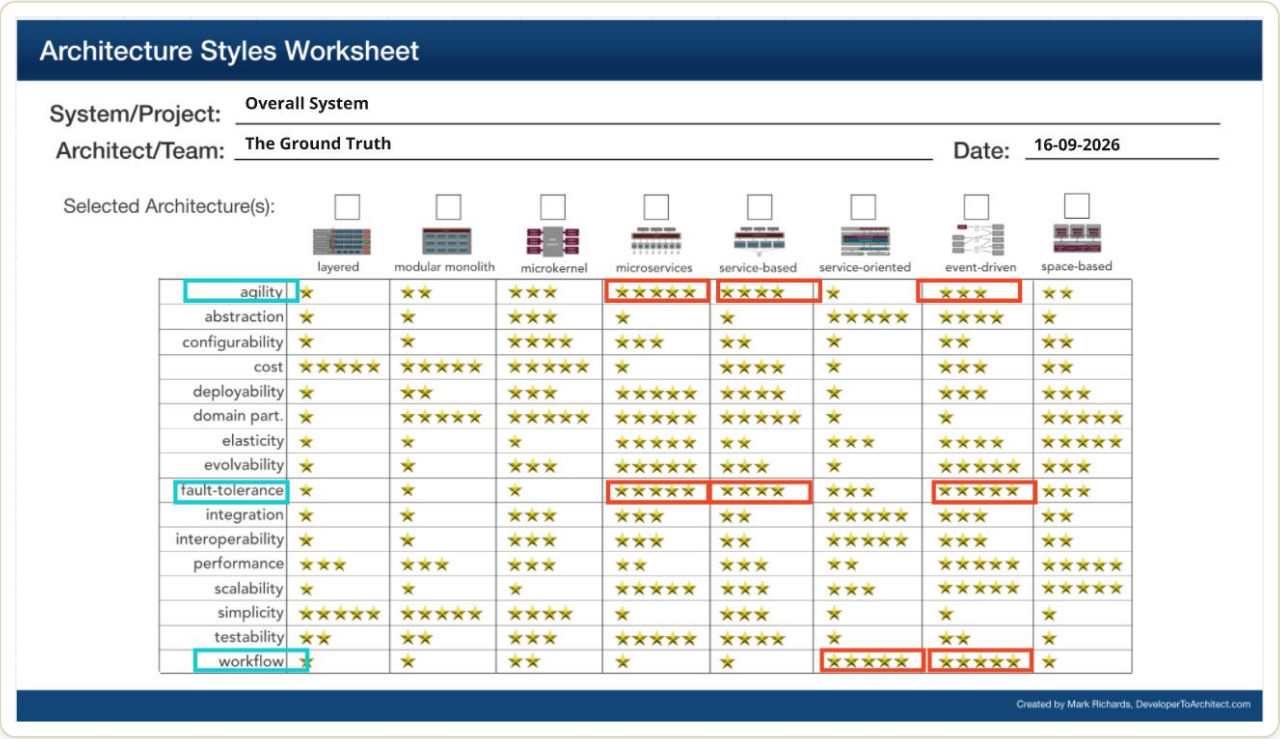
Fault tolerance, workflow, agility, availability, cost, safety latency

Added by AI

Accuracy, explainability, observability, provider portability, cost predictability

ARCHITECTURE STYLE

A service-based core, event-driven at the edge



Architecture Styles Worksheet

System/Project: Overall System

Architect/Team: The Ground Truth Date: 16-09-2026

Selected Architecture(s):

	layered	modular monolith	microkernel	microservices	service-based	service-oriented	event-driven	space-based
agility	★	★★	★★★	★★★★★	★★★★	★	★★★	★★
abstraction	★	★	★★★	★	★	★★★★★	★★★★	★
configurability	★	★	★★★★	★★★	★★	★	★★	★★
cost	★★★★★	★★★★★	★★★★★	★	★★★★	★	★★★	★★
deployability	★	★★	★★★	★★★★★	★★★★	★	★★★	★★★
domain part.	★	★★★★★	★★★★★	★★★★★	★★★★★	★	★	★★★★★
elasticity	★	★	★	★★★★★	★★	★★★	★★★★	★★★★★
evolvability	★	★	★★★	★★★★★	★★★	★	★★★★★	★★★
fault-tolerance	★	★	★	★★★★★	★★★★	★★★	★★★★★	★★★
integration	★	★	★★★	★★★	★★	★★★★★	★★★	★★
interoperability	★	★	★★★	★★★	★★	★★★★★	★★★	★★
performance	★★★	★★★	★★★	★★	★★★	★★	★★★★★	★★★★★
scalability	★	★	★	★★★★★	★★★	★★★	★★★★★	★★★★★
simplicity	★★★★★	★★★★★	★★★★	★	★★★	★	★	★
testability	★★	★★	★★★	★★★★★	★★★★	★	★★	★
workflow	★	★	★★	★	★	★★★★★	★★★★★	★

Created by Mark Richards, DeveloperToArchitect.com

Styles scored against our top three: agility, fault tolerance and workflow.

CHOSEN

Service-based core

Coarse-grained services per quantum own their data. Strong on agility and fault tolerance, and affordable for a small team.

Event-driven at the edge

MQTT store-and-forward between devices, edge and cloud. Covers workflow, where service-based alone scores one star.

WHY NOT

Microservices: lowest on cost and simplicity; too much to run for a small team

Space-based: extreme elasticity we don't need; telemetry scales with zones

Service-oriented: orchestration-heavy and weak on agility

Six quanta, each with its own characteristics

Ticket

User, Ticket, Payment and Visit services

Drives: Availability, recoverability

Estate

Plant, Ride, Staffing and Animal services

Drives: Fault tolerance, workflow, accuracy, explainability

Device Data

Device service: machines, sensors and readings

Drives: Workflow, fault tolerance

Job

Scheduler service for batch and timed work

Drives: Workflow

AI

Gateway, model registry and evaluators

Drives: Accuracy, explainability, observability

Notification

Notification and Report services

Drives: Availability, recoverability

Five use cases, one honest AI verdict each

Use case	AI verdict	What AI does, or why it doesn't	Decided in
1 Ticket and booking	**None, by design**	Signed tickets and capacity leases; there is nothing to predict	ADR-005, ADR-007
2 Animals and plants	**Enhancement**	Drafts keeper notes and summarizes evidence; never diagnoses	ADR-004, ADR-009
3 Footfall and popularity	**Enhancement**	Annotates a rule-based ranking with trends and anomalies	ADR-001, 002, 003
4 Feedback assessment	**Load-bearing**	Reads free text at 15,000 visitors a day; tags theme and zone	ADR-006, ADR-008
5 Growth and retention	**Enhancement**	Personalized itineraries once data exists; rules work on day one	README §7.5

ACROSS ALL FIVE



Ingestion of paper records



Alerts and notifications

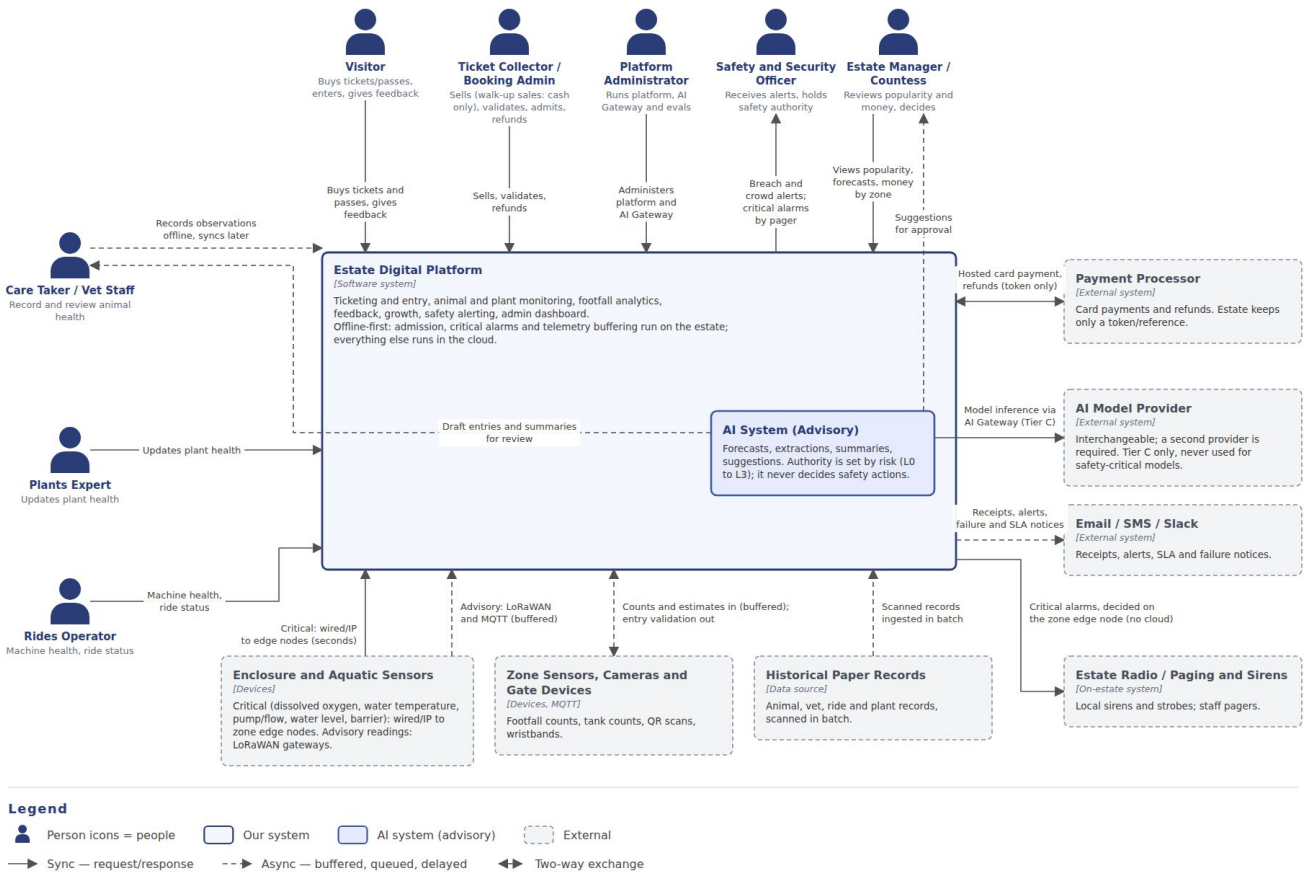


The Steward dashboard



AI quantum and Ledger

The whole system in one picture



Runs on the estate

Admission, critical alarms and telemetry buffering keep working when the uplink drops

Service-based core

Coarse-grained services per quantum; MQTT links devices, edge and cloud

AI advises, from inside

Authority set by risk, L0 to L3. Outside providers serve Tier C only, never safety

OFFLINE FIRST

A Wi-Fi drop slows analytics first, and safety never

ON THE ESTATE	ACROSS THE UPLINK	IN THE CLOUD
P0 Safety-critical — Critical alarms decided by rules on the zone edge node, in under 2 seconds	Never needs the uplink to decide	Logged afterwards for review
P1 Operational — Admission, telemetry and keeper records written at origin first	Forwarded when connected; marked stale while queued	Ticket, Estate and Animal services
P2 Analytical — Footfall batches and history	Batched; gaps marked, never invented	Reports, model training, the Ledger

Records are written durably at their origin before success is reported, then forwarded (ADR-004). Critical alarms never depend on the uplink (ADR-009).

Every failure, how we detect it, and what staff do

Failure View

What each failure does in each place, how we detect it, and what staff do until it is fixed. Read across a row.

	ON THE ESTATE — Keeps working when the uplink drops			OFF THE ESTATE — May pause; reports show how old their data is	
FAILURE · DETECTED BY	**DEVICE**	**ZONE EDGE NODE**	**GATE NETWORK**	**CLOUD**	**OPERATING RESTRICTION**
Internet uplink down Uplink monitor. The most common failure on this estate	Not affected	**Continues** Alarms sound locally. The broker buffers to disk; edge vision keeps counting	**Continues** Gates still reach the coordinator, so the shared log and the 7,500 ceiling hold	**Degrades** Online sales carry on; estate data arrives late and advisory concerns are marked historical. The backlog replays on return without double counting	**None** Staff see the age of the oldest unsent record ADR-004 · 005 · 009
A gate loses the admission coordinator The scanner cannot reach the coordinator	Not affected	Not affected	**Degrades** Scanners validate from their own cache (known duplicate window). The gate admits while its lease has balance and time, then stops	Not affected	**Supervisor at the gate** The gate escalates when its lease runs out. Its capacity is never reissued until the lease expires ADR-005 · 007
Zone edge node down Peer heartbeat loss; Sparkplug B death certificate in the cloud	**Degrades** Sensors keep sampling but nothing sounds the siren. Ride e-stops still halt rides locally	**Stops in that zone** Alarm rules, buffering and edge vision stop. A neighbouring node pages staff	Not affected	**Degrades** The zone is flagged on The Steward; its readings go stale	**Manual checks every 30 min** Critical tanks and enclosures checked by hand; paid animal encounters paused ADR-009
Power cut in a zone The UPS announces when it is on battery and when charge is low	**Continues** Critical sensors and sirens run on UPS	**Continues** Node and switch run on UPS and announce low charge before shutdown	Not affected	Not affected	**When the UPS runs low** As for "zone edge node down" ADR-009
Sensor or radio fault Stale-reading and probe-disagreement alarms	**That sensor stops** A silent critical sensor is an alarm, not "no news". Redundant oxygen probes cross-check	**Continues** Rules raise the stale-reading alarm and page staff	Not affected	**Degrades** If LoRaWAN drops, advisory readings are flagged stale	**Manual reading every 30 min** On that tank or enclosure until the sensor is replaced ADR-005 · 009
AI provider outage or price change Gateway errors; per-capability budget alert	Not affected	**Continues** Tier A edge models are self-hosted and never exposed to a provider	**Continues** Ticketing is AI-free	**Degrades** The gateway fails over to the second provider, or the Tier C feature pauses and feedback buffers	**None for safety** A capability that misses its bar on the new provider is demoted ADR-008 · 010

Design rule

No safety decision depends on the uplink, the cloud or an AI model. The one single point of failure we accept is the zone edge node: it is detected by its neighbour and covered by an operating restriction, and a zone moves to an independent alarm controller or paired nodes if the ADR-009 revisit triggers fire. Evidence status: **Designed.** Hardware drills are part of the ADR-009 assurance plan; nothing here is hardware-tested yet.

KEY

Continues · Degrades · Stops · Not affected · Operating restriction

----- Internet uplink: everything crossing it is buffered

Our top driving characteristic

Fault tolerance, failure by failure: device, edge node, gates and cloud

One owned single point of failure

The zone edge node. Its neighbour detects it; an operating restriction covers it

Evidence status: Designed

Hardware drills are planned in ADR-009. Nothing here is hardware-tested yet

KEY OUTCOMES

What the architecture commits to

Design guarantees

- Critical alarms in under 2 seconds, with no cloud and no AI in the path
- Admission and alarms survive at least 4 hours without the uplink
- The 7,500 safety ceiling holds even when gates lose contact
- Any Tier C AI provider swapped within a working day, by configuration
- Nothing that keeps a person or animal safe depends on a commercial AI provider

The ideas we'd defend first

1. Automate toward safety; only a person can relax it
2. Critical alarms are decided on the estate
3. Capacity leases hold the ceiling through a network split
4. The Ground Truth Ledger scores every AI capability
5. Popularity is a rule you can check by hand

A day at the estate

One Wednesday, two journeys, one platform. Every step shows how much the AI is allowed to do on its own.

INTERNET DOWN 09:05–09:40

VISITORS · THE WEBER FAMILY

Day before
Books a family pass in the app. The ticket is signed, so any gate can check it offline.

No AI

09:10
The gate scans the pass with the internet down. Admitted, inside the safety ceiling.

No AI

10:30
The app suggests Piranha Hall: a short wait, shown as a range and within crowd limits.

L0 · automatic

10:55
“No drinks near the piranhas.” AI tags the zone and routes the feedback to staff.

L0 tag · L1 route

11:00
Ops approves moving two food carts there, backed by the dwell data and the feedback.

L2 · person approves

Next month
A “You spoke, we acted” email, approved by marketing. The Webers come back as members.

L2 · person approves

KEEPER AYUSH · ANIMALS

07:40
Hesper has barely eaten. Ayush records a voice note in her enclosure, saved on the tablet.

No AI

09:12
Tank 12 oxygen falls. The edge node sounds the siren and pages Ayush in under two seconds.

No AI, no cloud

09:50
The tablet syncs. AI drafts fields from the voice note; Ayush corrects and confirms them.

L2 · person approves

09:55
Ayush opens a concern. AI links a similar note from 11 Sept. Paid encounters pause.

Advisory · L1 · pause

11:00
Dr Reddy examines Hesper. The diagnosis and treatment are hers alone.

L3 · person decides

Next day
Dr Reddy clears Hesper. Ayush resumes the paid encounters.

L2 · person approves

18:00
The Countess opens The Steward

- Admission and the Tank 12 alarm held through a 35-minute outage.
- Spend is up in Piranha Hall since the carts moved.
- Four encounters held and refunded. Welfare first.
- People made every call that mattered.
- Every AI action and its outcome is in the Ground Truth Ledger.

The Webers never knew the piranha tank was saved an hour before they arrived.

HOW MUCH THE AI MAY DO ALONE — No AI by design · L0 acts, sampled for review · L1 acts, notifies, undoable · L2 a person approves · L3 a person decides · Advisory explains, never acts

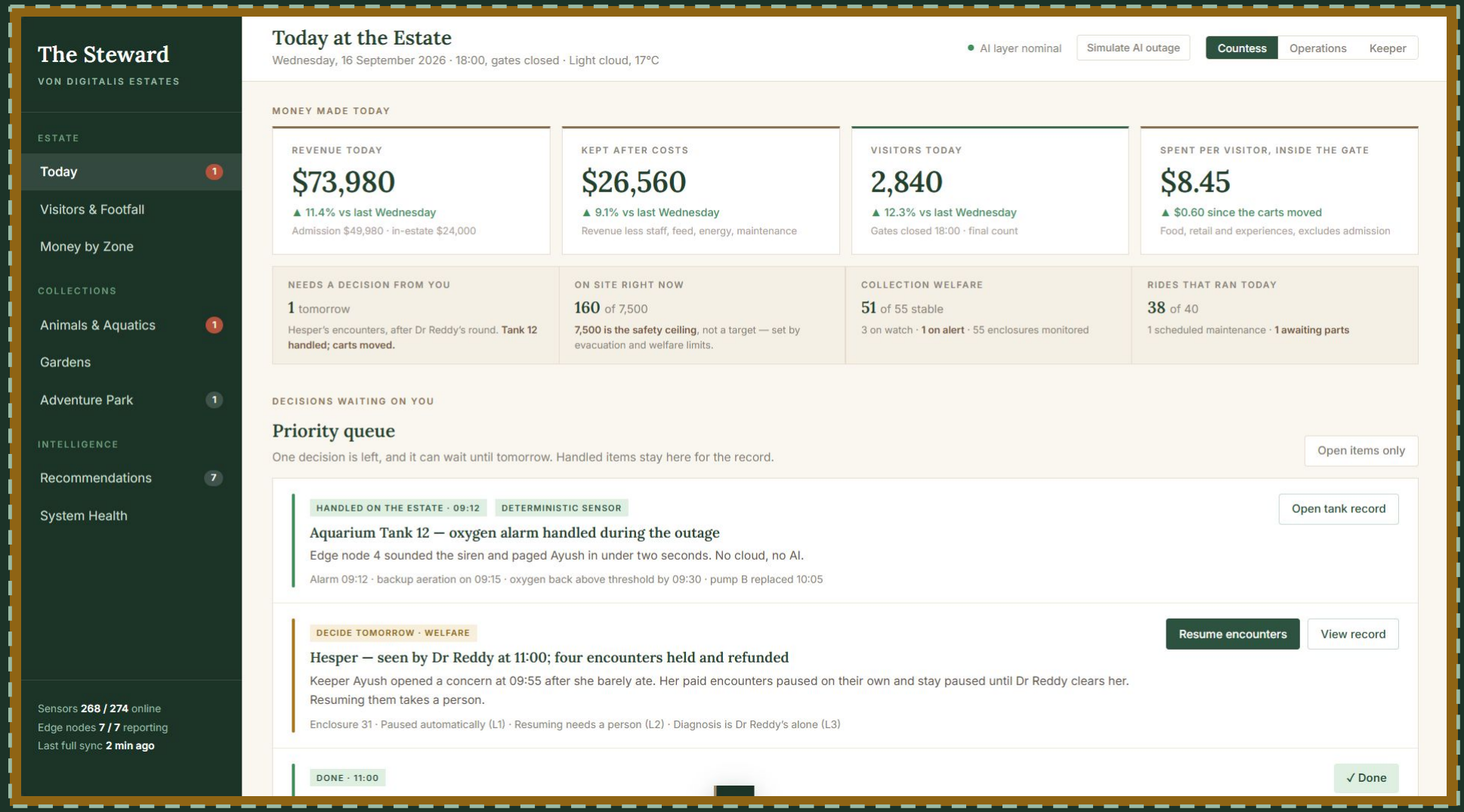
Times match The Steward prototype. Offline admission: ADR-005, ADR-007 · Critical alarms on the estate: ADR-009 · Feedback routing: ADR-006 · Authority by risk: ADR-010

Principle: automate toward safety; only a person can relax it. The system paused Hesper’s encounters on its own. Resuming them took a person.

The Steward

The Countess's view at six o'clock: money first, then only the decisions that need her.

- Decisions waiting on you
- Why each suggestion was made
- Countess, operations and keeper views
- AI outages announced, never silent



Live: santhosh-challa.github.io/ground-truth

Ticketing: AI-free, and built to survive outages

AI VERDICT · NONE, BY DESIGN

Decided in ADR-005 (deployment topology) and ADR-007 (ticketing system)

How it works

One Ticket Service, two channels

Online and counter sales share prices, passes and validity rules

A signed ticket

Any gate verifies authenticity offline, with no network call

An admission coordinator on the estate

Tracks occupancy and keeps the shared admission log

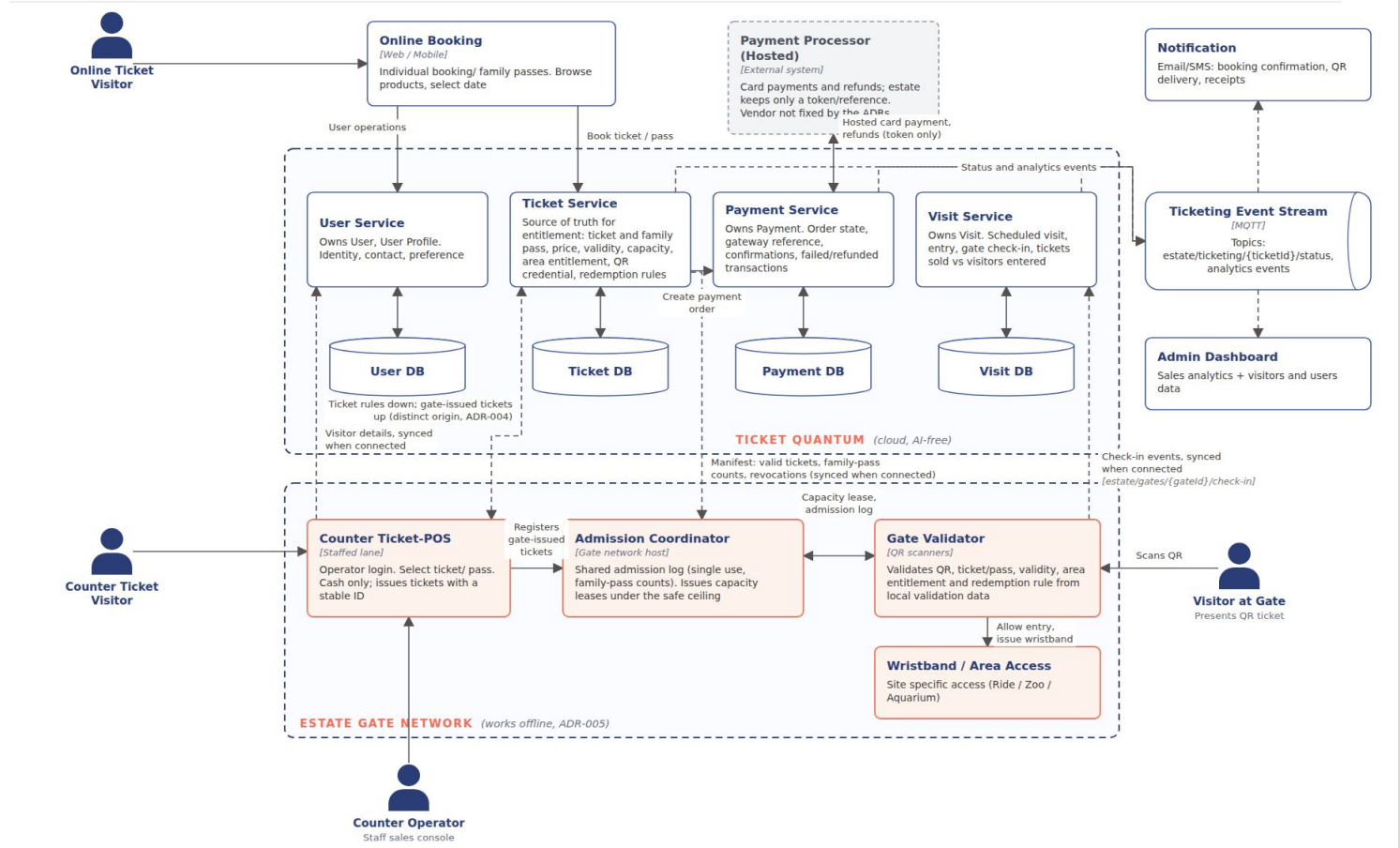
Capacity leases

Each gate admits only its time-boxed share of the 7,500 ceiling; an unreachable gate's share is never reissued

Trade-off	How we contain it
A copied QR still verifies	The admission log enforces single use
Offline gates may hold stale data	Controlled cache validity; resync on reconnect
Duplicate entry during an outage	Unique event IDs, a bounded window, reconciliation
Coordinator and gate network are single points	Local scanner logs; supervisor escalation
Payment and ticket state can diverge	Idempotency and payment-webhook reconciliation
Card payments need the uplink	Walk-up sales on the estate are cash only

Rejected: ticket rules in every channel (duplicated logic, pricing drift); cloud only with device caches (no cross-gate duplicate check, no alarms in an outage).

Ticketing containers



Reading the C2

Four services own their aggregates: User, Ticket, Payment and Visit. Online booking and the counter share one Ticket Service (ADR-007).

Animals: keepers and rules raise concerns, AI assists

AI VERDICT · ENHANCEMENT

Decided in ADR-004 (record durability and sync) and ADR-009 (critical alarms at the edge)

Three flows

Recording an observation

Note, voice or photo saved on the tablet first, synced later with the same ID. AI drafts the fields; the keeper confirms (L2).

Reviewing a concern

Configured thresholds raise concerns with no AI. An optional summary links the evidence; it never diagnoses.

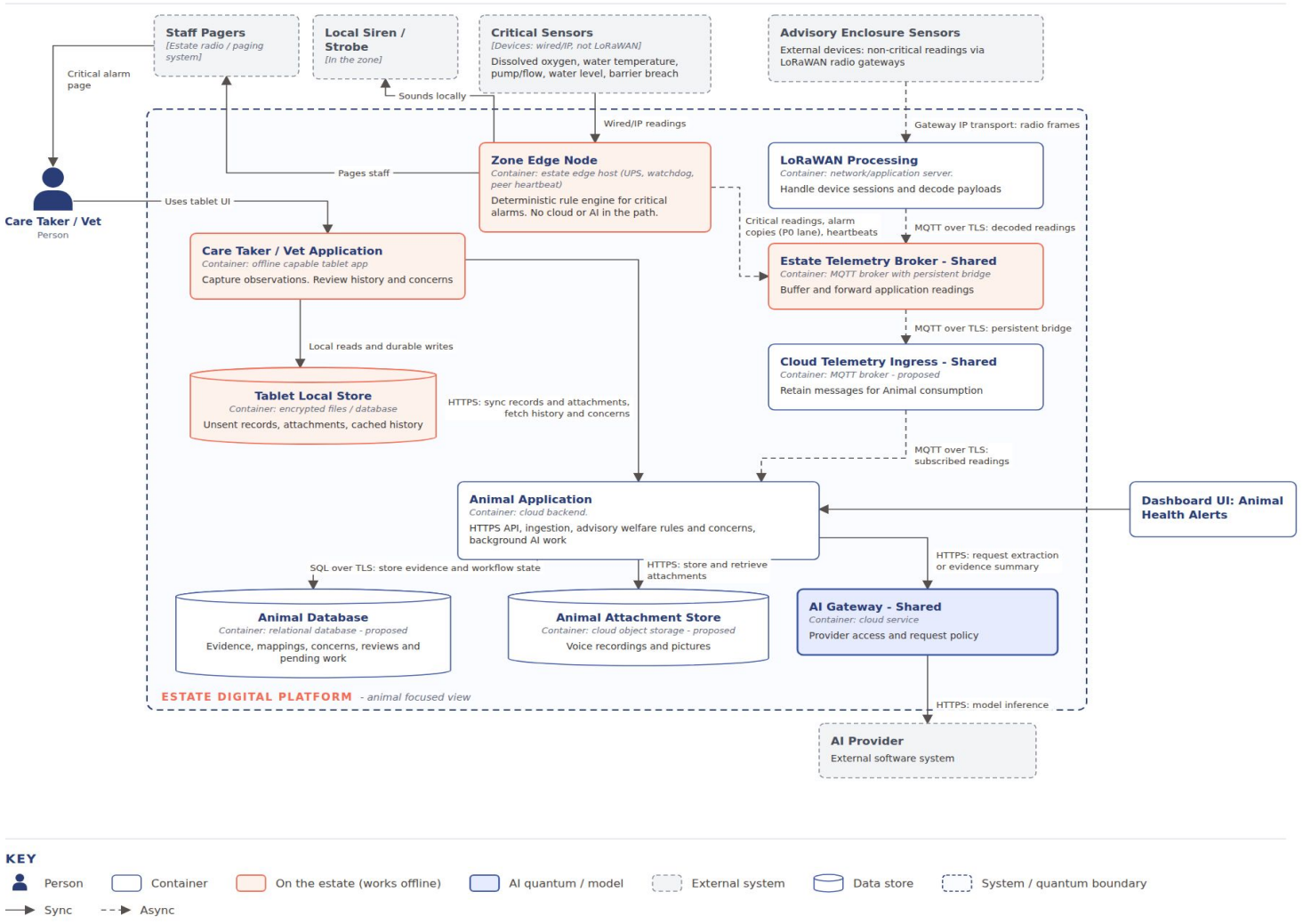
Counting the piranhas

A camera pilot reports estimates with view quality, checked against keeper counts. Never a census.

Trade-off	How we contain it
Every origin needs storage and sync	Loss is bounded; backlog metrics are a live signal
Cloud reports lag reality	Late readings marked historical, never shown as current
No illness classifier	We'd never have the labelled data; keepers and thresholds instead
Advisory concerns can tire keepers	Precision-first, with an alert budget; critical alarms exempt
Plants share the pattern	Same broker and rules, owned by the horticulturist

Rejected: sending everything straight to the cloud and accepting loss. With patchy Wi-Fi, loss would be routine.

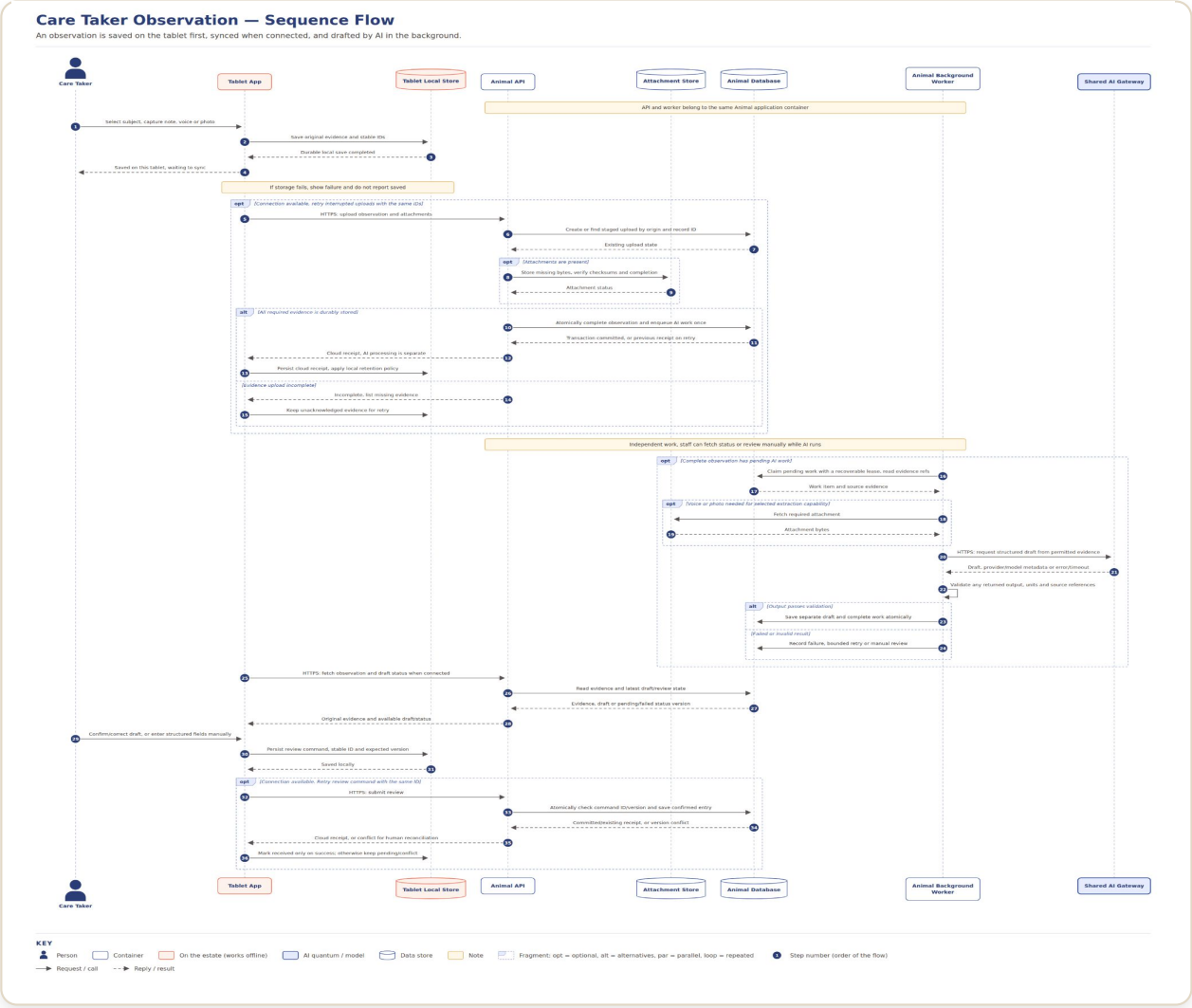
Animal monitoring containers



Reading the C2

- The tablet writes locally before anything is reported as saved
- Advisory readings travel LoRaWAN, broker, cloud, buffered at each hop
- The Animal application evaluates configured rules and keeps the evidence
- The shared AI gateway drafts fields and summaries; advisory only
- Critical alarms take their own path on the zone edge node

How an observation survives a dead signal



1. Saved on the tablet with a stable ID before the keeper is told it is saved
2. Uploads retry with the same IDs, so nothing is counted twice
3. AI work is queued only once all the evidence has arrived
4. AI drafts fields from permitted evidence; output is validated or retried
5. The keeper corrects and confirms; version conflicts go to a person

Critical alarms are decided on the estate

Class	Examples	Tolerable delay	Who decides
Critical alarm	Oxygen, water temperature, pump or flow, water level, barrier breach, emergency stop	Seconds	Deterministic rules on the edge node
Advisory concern	Thresholds on non-critical readings, keeper observations	Hours	A keeper or vet, in the cloud
AI evidence summary	Readings and notes attached to a concern	Can pause	Advisory only; never needed to act

Options we weighed

Cloud rules, with keeper rounds as backup	REJECTED
Rules on the edge node, with UPS, watchdog and peers	CHOSEN
A controller per critical point	IF A NODE FAILS
Paired active and standby nodes	IF A NODE FAILS

Containing the single point of failure

- UPS and a watchdog on every alarm node
- Peer nodes detect a dead neighbour and page staff
- Weekly siren tests; monthly power-removal drills
- A zone opens only after its alarm passes on site

AI cameras may add alerts; never suppress or delay one.

Popularity is a rule anyone can check

AI VERDICT · ENHANCEMENT

Decided in ADR-001 (scoring), ADR-002 (idempotent reconciliation) and ADR-003 (drift monitoring)

ADR-001 **Hybrid, phased**

Launch: a rule-based score from footfall and dwell time. Later: forecasts and anomaly flags annotate it, never replace it.

ADR-002 **Idempotent and reconciled**

Writes keyed by device, zone and window. Counts reconciled before scoring; mismatches go to a dead-letter queue.

ADR-003 **Drift with a human gate**

Forecasts scored every batch. On a breach, ML notes are hidden and an engineer reviews.

Trade-off	How we contain it
No footfall history on day one	Phase 1 runs on rules alone
Reconciliation adds latency	Negligible against hourly and nightly batches
Late data looks like a gap	Replayed with its true timestamp, never double-counted
A sensitive drift threshold hides real trends	A conservative start, tuned through the review loop

Dwell time beats headcount: a busy corridor is not a popular exhibit.

Rejected: pure ML scoring (no data on day one, hard to explain); best-effort counting; after-the-fact audits.

USE CASE 4 · FEEDBACK

Feedback: where AI is load-bearing, and fenced in

AI VERDICT · LOAD-BEARING

Decided in ADR-006 (feedback evaluation and reporting) and ADR-008 (AI quantum)

Collect
App or kiosk; queued offline

→

Sanitise
Personal data stripped before queueing

→

Evaluate
AI tags sentiment, theme and zone (L0)

→

Route
To the zone manager (L1)

→

Report
Deterministic; never calls a model

Urgent safety or welfare reports skip AI entirely: keyword rules and an "urgent" option route them straight to zone staff.

Requirement	Async, behind a gateway (chosen)	Direct LLM call (rejected)
Works offline	High: buffered at the edge	Low
Provider swap	A configuration change	A code rewrite
Personal data	Stripped before buffering	Pass-through risk
Report reliability	Reads a structured database	Live hallucination risk
Complexity	Higher	Lower

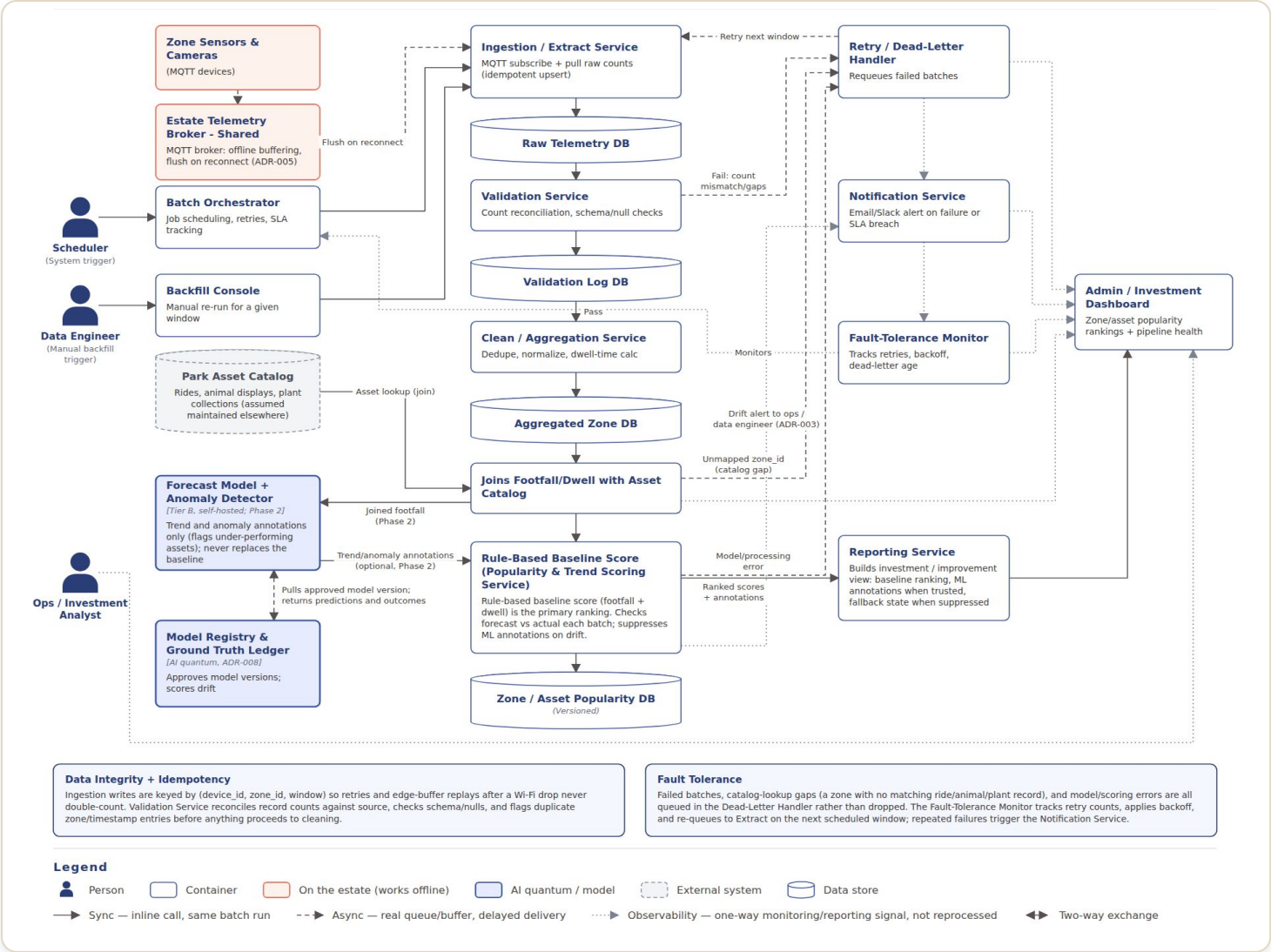
Trade-offs we accepted

Insights are not instant: hourly reports

Models return malformed output: strict schema checks at the gateway

The gateway is a dependency: high availability; feedback waits in the buffer

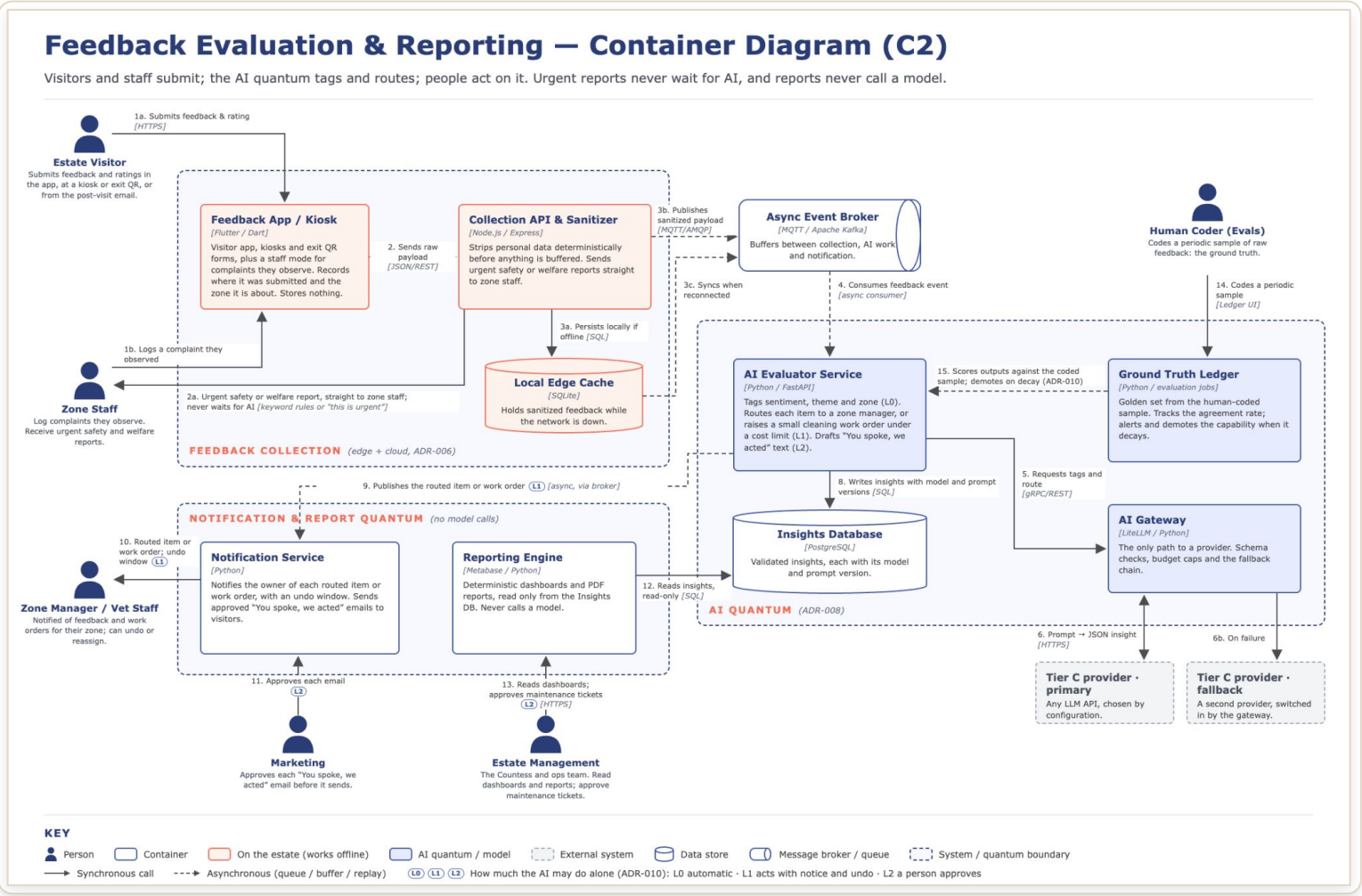
Zone popularity containers



Reading the C2

- Edge gateways buffer readings and flush on reconnect
- Ingestion upserts by device, zone and window
- Validation reconciles counts before cleaning
- Aggregation computes dwell time per zone
- Failures retry, then dead-letter, with alerts
- The dashboard ranks zones and assets for investment

Feedback containers



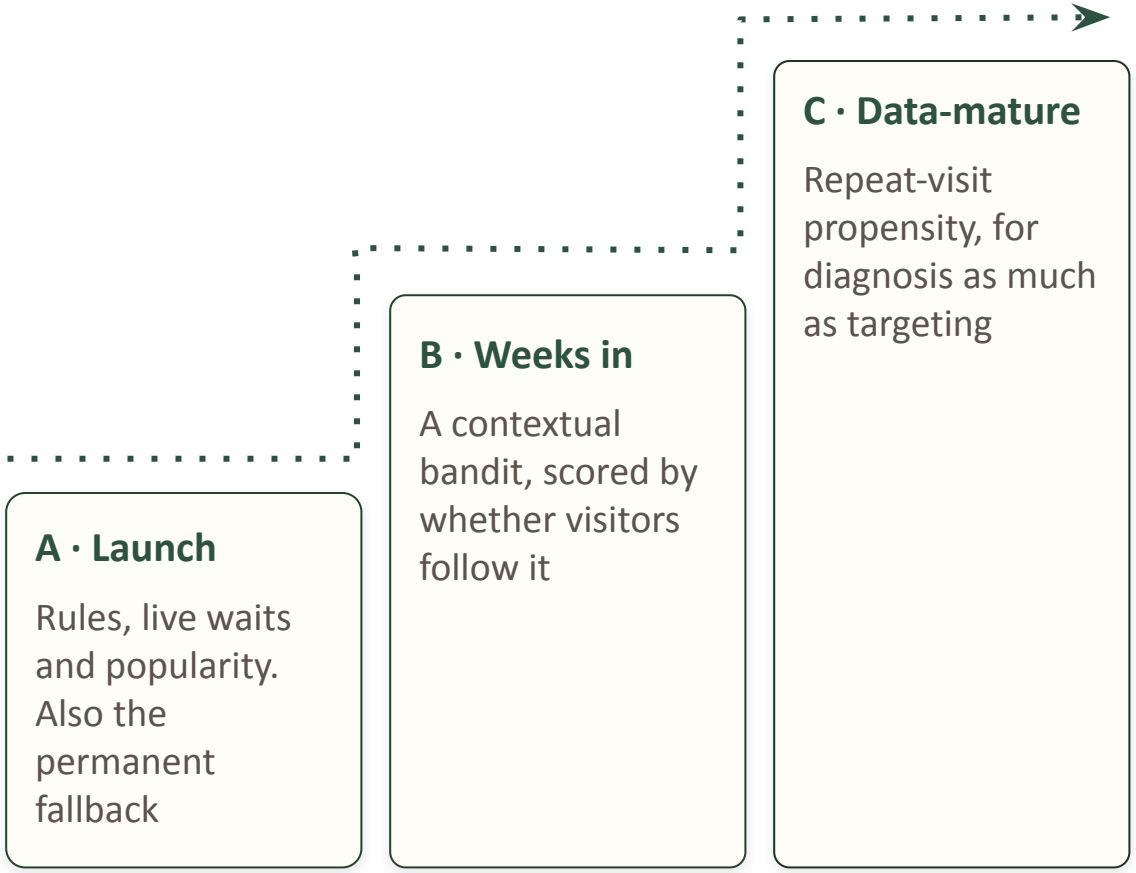
Reading the C2

- Visitors submit through the app or a kiosk, which queues while offline
- Personal data is stripped first. Urgent reports go straight to zone staff published
- AI tags each item (L0), then routes it to a zone manager or raises a small work order, with an undo window (L1)
- The Ledger scores tagging against a human-coded sample and demotes it if agreement drops
- Marketing approves every “You spoke, we acted” email. Managers approve maintenance tickets (L2)
- Reports read only from the Insights DB and never call a model

Growth: rules on day one, learning once there's data

AI VERDICT · ENHANCEMENT

Designed at the level of the README (section 7.5)



One intervention, three outcomes

A wait-aware itinerary in the visitor app suggests what to see next from live congestion.

- **Visitors queue less**
- **Crowds spread away from pinch points**
- **Crush risk falls**

Measured, not assumed

Held-out control groups compare return rate, dwell time and in-estate spend. Consent is held separately, and only visit behaviour is used.

ACROSS ALL FIVE

Ingestion, alerts and The Steward



Ingestion of paper records

AI READS, A PERSON CHECKS

- Decades of animal, vet, ride and plant records exist only on paper
- Confidence-scored extraction; uncertain fields go to a person
- The scan stays the source of truth; gaps stay gaps
- Batch work on the analytical lane, in Phase 2



Alerts and notifications

NO AI IN ROUTING

- Severity decides the route and the channel
- Alert budgets per role for advisory concerns only
- Every alert closed with a verdict that feeds the Ledger
- Critical alarms sound from the edge node



The Steward dashboard

SHOWS AI, DECIDES NOTHING

- Money first, then the decisions waiting on you
- Why each suggestion was made, with its model version
- Views for the Countess, operations and keepers
- Announces AI degradation instead of going quiet

RISK-BASED AI STRATEGY

AI authority is set by risk, not by model confidence

Who may supply the model?

TIER A — **Safety and counting**

Self-hosted on edge nodes. Never a provider.

TIER B — **Forecasts and suggestions**

Self-hosted, with a rule-based fallback.

TIER C — **Feedback and keeper notes**

A provider is allowed, only through the gateway.

What may it do on its own?

L0 — **Acts automatically, sampled for review**

Wait-time ranges, popularity notes, feedback tags

L1 — **Acts, notifies, can be undone**

Pausing an encounter, routing feedback

L2 — **A person approves**

Keeper-note fields, resuming an encounter, emails

L3 — **Only a person, ever**

Lockdown, evacuation, all-clear, diagnosis and treatment

How authority changes: *Start at L2 → shadow mode → promoted on evidence, owner approves → auto-demoted by the Ledger → L3 never moves (ADR-010)*

Automate toward safety. Only a person can relax it.

Set by five factors: harm if wrong, reversibility, measured performance, input health, response capacity.

One AI quantum absorbs the churn of the AI landscape

Inside the AI quantum

Gateway
The only path to an outside provider; routing and fallback

Model registry
Approved versions; edge models pull from it

Evaluators
A fixed evaluation set per capability

Budget caps
Per-capability spend limits and early alerts

Versioned prompts
Prompts and thresholds under change control

A better model appears	Swap it by configuration; contract tests on a second provider prove it
The price rises	Budget caps alert long before the invoice does
A provider shuts down	Fall back; nothing safety-critical ever depended on it

Options we weighed (ADR-008)

Each domain calls providers directly	REJECTED
A gateway quantum, AI inside each domain	REPEATS EVALS
One AI quantum, with split triggers	CHOSEN

The Ground Truth Ledger scores every capability

1 Golden sets
Versioned, per capability

→

2 Evaluation gate
Every change must pass before release

→

3 Shadow and canary
Runs silently, rolls back on regression

→

4 Production monitors
Drift, confidence, cost, silent failure

→

5 Human verdicts
Each alert closed real, false or unsure

Every verdict becomes a label that grows the golden sets.

Capability	Ground truth	Posture	Demoted when
Wait-time prediction	Actual wait at the queue front	Error bands, not points	Rolling error breaches
Footfall forecast	Gate count at close of day	Must beat a seasonal baseline	It fails the baseline
Feedback tagging	A human-coded sample	Agreement rate	Agreement decays
Keeper-note extraction	The keeper's confirmed entry	Keeper confirms every draft	Corrections rise
Piranha count	A weekly manual count	A range, never a point	The interval widens
Safety camera detection	Drills and shadow mode	Recall first	Drill recall drops

Every decision names the check that enforces it

Fitness functions for our top three

Fault tolerance

Four-hour uplink chaos test: admission and alarms continue; the backlog replays

Workflow

Replay duplicated, retried and late events; records stay unchanged

Agility

No component outside the AI quantum depends on a provider

Safety latency

Synthetic trip injection: an alarm in under 2 seconds

ADR check	What it commits to	ADR	Status
`ai_free_ticketing`	Nothing in ticketing depends on AI	007	**Planned**
`critical_alarm_local`	No critical alarm passes through the cloud	009	**Planned**
`no_direct_provider_calls`	Only the AI quantum calls a provider	008	**Planned**
`authority_matches_risk`	High-harm actions stay at L2 or L3	010	**Planned**
`baseline_primary`	ML annotates popularity, never replaces it	001	**Planned**
`idempotent_replay`	Retries never duplicate a record	002, 004	**Planned**

Specified and planned against a declared architecture model, so they can run in CI from the first build. We don't claim a check runs before it does.

What could go wrong, and what we left out on purpose

Risk	Severity	Mitigation built in
Safety event missed by the AI layer	**Critical**	The deterministic layer works alone; AI tuned for recall
Long outage on a peak day	**High**	Edge admission, leases and local alarms; a 4-hour chaos test
Model quality drifts after launch	**High**	Evaluation gate, canary rollback, automatic demotion
Provider price rise or shutdown	**Medium**	The AI quantum, tiering and budget caps
Paper records corrupted in ingestion	**High**	Confidence scoring, human review, the scan kept
Growth outruns safe capacity	**Critical**	Leases set by safety limits, never by demand

Left out on purpose

- ✕ **AI in ticketing**
  nothing to predict
- ✕ **AI ordering a safety action**
  lockdown, evacuation and all-clear stay with a person
- ✕ **An illness classifier**
  the labelled data would never exist
- ✕ **A single piranha number**
  a range is the honest answer
- ✕ **Facial recognition**
  we count people, we never identify them

ROADMAP

Three phases, each with exit criteria

PHASE 1 · Q1 TO Q2

The safe, offline core

Ticketing, footfall, animal monitoring, safety and feedback, with the AI quantum and Ledger in place first

Exit: alarms tested in every zone, a four-hour outage survived, every AI capability through shadow mode

PHASE 2 · Q3

Earned intelligence

Paper records ingested. Capabilities promoted from L2 toward L1 only on measured performance

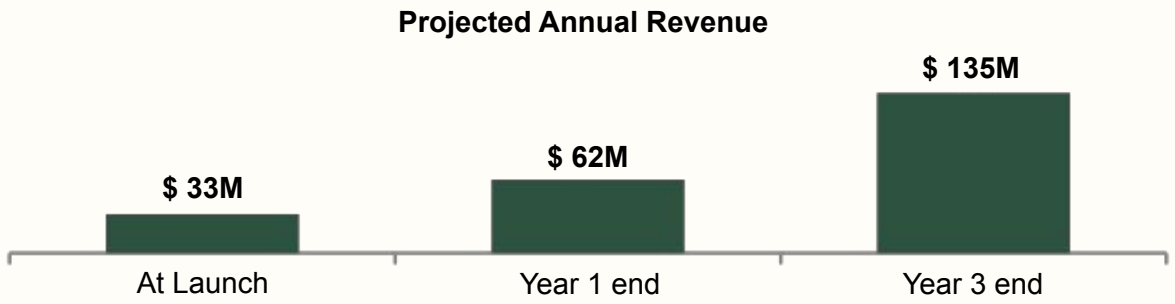
Exit: forecasts beat a seasonal baseline before they switch on

PHASE 3 · Q4 ONWARD

Growth surfaces

A visitor assistant and yield-managed pricing

Exit: a year of demand data first, and a person approves every price change



Visitors a day: 5,000 → 8,000 → 15,000

Peak on site: 3,000 → 4,500 → 7,000, under the 7,500 ceiling

Revenue per visitor: $22 → $26 → $30

Projections from assumptions stated in our README.

THANK YOU

Every model here is optional, replaceable and continuously scored.

THREE THINGS WE LEARNED

We kept wanting to add AI. Most of the work was taking it back out.

Designing for a network you can't trust changes where decisions live, not just how you retry.

The hard question was never which model, but how we'd know it still works in three months.

Team Ground Truth

Repository: github.com/santhosh-challa/ground-truth
Live prototype: santhosh-challa.github.io/ground-truth